CBSE Mathematics 2015

Class X and Class XII



Working solutions and source audit

About this working file

This project follows the folder and LaTeX conventions used in the EE1010 course folder: questions and solutions are stored in `chapters/`; plotting scripts and the `CoordGeo` helper package are under `codes/`; generated figures are under `figs/`; and the root document imports the style files `gvv-book.sty` and `gvv.sty`. The figures included here are produced by Python scripts, not drawn manually inside the LaTeX questions.

Audit status. The current batch includes Class X questions 1–4 from Section A, question 5 from Section B, and Class XII Vectors questions 1–7 from the cited public transcriptions. Class X question 2 has been checked against the official marking scheme. Class XII vector questions 1–2 retain source-wording flags and remain conditional. The Class X chord/tangent figure has been redrawn with the official point placement and its angle derived accordingly.

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1. Class X Mathematics

A. Section A

1.1.1 If the quadratic equation $px^2 - 2\sqrt{5}px + 15 = 0$ has two equal roots, find the value of p. (CBSE Class X, 2015, Q1)

Solution: For a quadratic equation $ax^2 + bx + c = 0$ to have equal roots, its discriminant must vanish:

$$b^2 - 4ac = 0. \tag{1.1.1.1}$$

Here $a = p$, $b = -2\sqrt{5}p$ and $c = 15$. Therefore,

$$(-2\sqrt{5}p)^2 - 4(p)(15) = 0 \tag{1.1.1.2}$$

$$20p^2 - 60p = 0 \tag{1.1.1.3}$$

$$20p(p-3) = 0. \tag{1.1.1.4}$$

The algebraic possibilities are $p = 0$ or $p = 3$. However, $p = 0$ removes the quadratic term and leaves $15 = 0$, which has no solution. Hence

$$\boxed{p = 3}. \tag{1.1.1.5}$$

1.1.2 In Figure 1.1.2.1, a tower AB is $20\,\mathrm{m}$ high and BC, its shadow on the ground, is $20\sqrt{3}\,\mathrm{m}$ long. Find the Sun's altitude. (CBSE Class X, 2015, Q2)

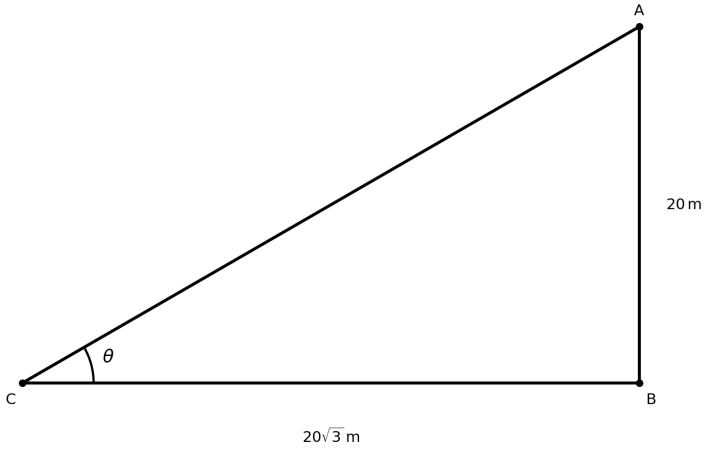


Fig. 1.1.2.1: Tower and shadow; dimensions match the marking scheme.

Solution: The angle of elevation of the Sun is θ. In right triangle ABC,

$$\tan\theta = \frac{AB}{BC} = \frac{20}{20\sqrt{3}} = \frac{1}{\sqrt{3}}, \ \therefore \quad \boxed{\theta = 30^\circ}. \tag{1.1.2.1}$$

The length $20\sqrt{3}\,\mathrm{m}$ is confirmed by the CBSE 2015 Class X marking scheme. The earlier source transcription omitted the radical and multiplier.
The figure was generated by the following Python script:

codes/Tower.py

1.1.3 Two different dice are tossed together. Find the probability that the product of the two numbers on the top of the dice is 6. (CBSE Class X, 2015, Q3)
Solution: The two dice produce $6 \times 6 = 36$ equally likely ordered outcomes. The favourable ordered pairs are

$$(1,6),\ (6,1),\ (2,3),\ (3,2). \tag{1.1.3.1}$$

There are four favourable outcomes. Thus,

$$P(\text{product} = 6) = \frac{4}{36} = \boxed{\frac{1}{9}}. \tag{1.1.3.2}$$

1.1.4 In Figure 1.1.4.1, PQ is a chord of a circle with centre O and PT is a tangent. If $\angle QPT = 60^\circ$, find $\angle PRQ$. (CBSE Class X, 2015, Q4)

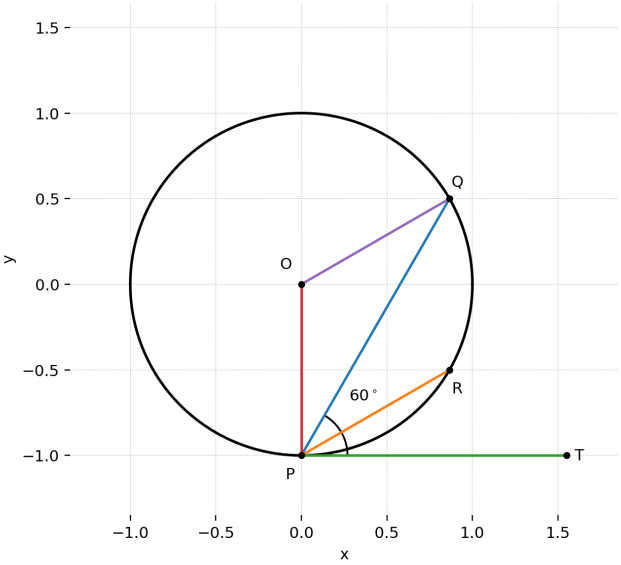


Fig. 1.1.4.1: Chord and tangent configuration; drawn by Python.

Solution: Since a radius is perpendicular to the tangent at the point of contact, $\angle OPT = 90^\circ$. Hence

$$\angle OPQ = \angle OPT - \angle QPT = 90^\circ - 60^\circ = 30^\circ. \tag{1.1.4.1}$$

As $OP = OQ$, triangle OPQ is isosceles, so $\angle OQP = 30^\circ$ and

$$\angle POQ = 180^\circ - 30^\circ - 30^\circ = 120^\circ. \tag{1.1.4.2}$$

The point R lies on the minor arc PQ in the given figure. Therefore $\angle PRQ$ subtends the major arc PQ, whose measure is $360^\circ - 120^\circ = 240^\circ$. Thus

$$\angle PRQ = \frac{240^\circ}{2} = \boxed{120^\circ}. \tag{1.1.4.3}$$

The figure was generated by the following Python script:

codes/TangentChord.py

B. Section B

1.2.5 Two tangents RQ and RP are drawn from an external point R to a circle with centre O. If $\angle PRQ = 120^\circ$, prove that $OR = PR + RQ$. (CBSE Class X, 2015, Q5)

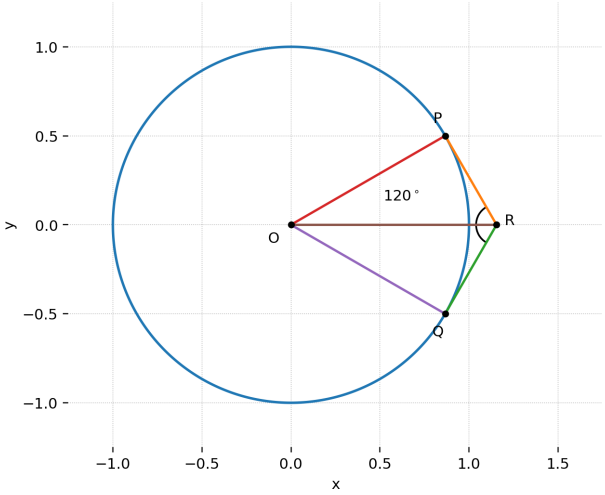


Fig. 1.2.5.1: Two tangents from an external point; drawn by Python.

Solution: The radii to the points of contact are perpendicular to the tangents: $OP \perp PR$ and $OQ \perp QR$. The right triangles $\triangle OPR$ and $\triangle OQR$ are congruent by RHS, since $OP = OQ$ and OR is common. Hence

$$PR = RQ, \qquad \angle PRO = \frac{1}{2}\angle PRQ = 60^\circ. \tag{1.2.5.1}$$

In right triangle OPR,

$$\cos 60^\circ = \frac{PR}{OR} \tag{1.2.5.2}$$

$$\frac{1}{2} = \frac{PR}{OR} \tag{1.2.5.3}$$

$$OR = 2PR. \tag{1.2.5.4}$$

Using $PR = RQ$ gives

$$\boxed{OR = PR + RQ}. \tag{1.2.5.5}$$

The figure was generated by the following Python script:

codes/TwoTangents.py

C. Batch audit

Questions 1–5 in this batch are now based on the official 2015 Class X wording and figure configuration. The chapter ordering separates Question 5 into Section B, matching the official paper. Continue auditing the remaining questions against the official paper before submission.

2. CLASS XII MATHEMATICS

A. Vectors

2.1.1 Find a vector of magnitude $\sqrt{171}$ which is "to both" of the vectors $\mathbf{a} = \hat{i} - 2\hat{j} - 3\hat{k}$ and $\mathbf{b} = 3\hat{i} - \hat{j} - 2\hat{k}$. (Source wording retained)

Solution: Wording flag. The source sentence omits the relation intended by "to both". The following result assumes it means "perpendicular to both". A vector perpendicular to both $\mathbf{a}$ and $\mathbf{b}$ is parallel to their cross product:

$$\mathbf{a} \times \mathbf{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -2 & -3 \\ 3 & -1 & -2 \end{vmatrix} \tag{2.1.1.1}$$

$$= \hat{i} - 7\hat{j} + 5\hat{k}. \tag{2.1.1.2}$$

Its magnitude is $\sqrt{1 + 49 + 25} = \sqrt{75}$. Multiplying by $\sqrt{171}/\sqrt{75} = \sqrt{57}/5$ gives the required magnitude. Under the perpendicular interpretation,

$$\boxed{\mathbf{v} = \pm\frac{\sqrt{57}}{5}\left(\hat{i} - 7\hat{j} + 5\hat{k}\right)}. \tag{2.1.1.3}$$

Confirm the missing word against the original source before final submission.

2.1.2 In triangle OAC, B is the midpoint of AC. The source then states $\overrightarrow{OA} = \mathbf{a}$, $\overrightarrow{OA} = \mathbf{b}$ and asks for $\overrightarrow{OC}$. (Source wording retained)

Solution: Wording flag. The source repeats $\overrightarrow{OA}$ in the second given vector; taken literally, the information does not determine $\overrightarrow{OC}$. If the second vector was intended to be $\overrightarrow{OB} = \mathbf{b}$, then the midpoint relation gives

$$\overrightarrow{OB} = \frac{\overrightarrow{OA} + \overrightarrow{OC}}{2}, \tag{2.1.2.1}$$

$$\mathbf{b} = \frac{\mathbf{a} + \overrightarrow{OC}}{2}, \tag{2.1.2.2}$$

$$\boxed{\overrightarrow{OC} = 2\mathbf{b} - \mathbf{a}}. \tag{2.1.2.3}$$

This is a conditional answer only; verify the vector named in the source.

2.1.3 If $|\mathbf{a}| = a$, find

$$|\mathbf{a} \times \hat{i}|^2 + |\mathbf{a} \times \hat{j}|^2 + |\mathbf{a} \times \hat{k}|^2.$$

Solution: Write $\mathbf{a} = a_x\hat{i} + a_y\hat{j} + a_z\hat{k}$. Then

$$|\mathbf{a} \times \hat{i}|^2 = a_y^2 + a_z^2, \tag{2.1.3.1}$$

$$|\mathbf{a} \times \hat{j}|^2 = a_x^2 + a_z^2, \tag{2.1.3.2}$$

$$|\mathbf{a} \times \hat{k}|^2 = a_x^2 + a_y^2. \tag{2.1.3.3}$$

Adding these expressions gives

$$2(a_x^2 + a_y^2 + a_z^2) = 2|\mathbf{a}|^2 = \boxed{2a^2}. \tag{2.1.3.4}$$

2.1.4 The vectors $\mathbf{a} = 3\hat{i}+x\hat{j}$ and $\mathbf{b} = 2\hat{i}+\hat{j}+y\hat{k}$ are mutually perpendicular. If $|\mathbf{a}| = |\mathbf{b}|$, find y.

Solution: Perpendicularity gives

$$\mathbf{a}\cdot\mathbf{b} = 6 + x = 0 \quad \Rightarrow \quad x = -6. \tag{2.1.4.1}$$

Equality of magnitudes then gives

$$9 + x^2 = 2^2 + 1^2 + y^2, \tag{2.1.4.2}$$
$$9 + 36 = 5 + y^2, \tag{2.1.4.3}$$
$$y^2 = 40. \tag{2.1.4.4}$$

Therefore, both real values are

$$\boxed{y = \pm 2\sqrt{10}}. \tag{2.1.4.5}$$

2.1.5 Let $\mathbf{a} = \hat{i} + 4\hat{j} + 2\hat{k}$, $\mathbf{b} = 3\hat{i} - 2\hat{j} + 7\hat{k}$ and $\mathbf{c} = 2\hat{i} - \hat{j} + 4\hat{k}$. Find $\mathbf{d}$ perpendicular to both $\mathbf{a}$ and $\mathbf{b}$ such that $\mathbf{c}\cdot\mathbf{d} = 27$.

Solution: A vector perpendicular to both $\mathbf{a}$ and $\mathbf{b}$ is proportional to $\mathbf{a}\times\mathbf{b}$:

$$\mathbf{a}\times\mathbf{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 4 & 2 \\ 3 & -2 & 7 \end{vmatrix} = 32\hat{i} - \hat{j} - 14\hat{k}. \tag{2.1.5.1}$$

Since $\mathbf{c}\cdot(\mathbf{a}\times\mathbf{b}) = 2(32) + (-1)(-1) + 4(-14) = 9$, choosing the scale factor 3 yields the required dot product. Thus

$$\boxed{\mathbf{d} = 96\hat{i} - 3\hat{j} - 42\hat{k}}. \tag{2.1.5.2}$$

2.1.6 Find the shortest distance between the lines

$$\mathbf{r} = (\hat{i} + 2\hat{j} + 3\hat{k}) + \lambda(2\hat{i} + 3\hat{j} + 4\hat{k}),$$
$$\mathbf{r} = (2\hat{i} + 4\hat{j} + 5\hat{k}) + \mu(4\hat{i} + 6\hat{j} + 8\hat{k}).$$

Solution: The direction vectors are proportional, so the lines are parallel. Let $\mathbf{a} = (1,2,3)$ be a point on the first line, $\mathbf{b} = (2,4,5)$ be a point on the second, and $\mathbf{v} = (2,3,4)$ be their common direction. The distance between parallel lines is

$$d = \frac{|(\mathbf{b}-\mathbf{a})\times\mathbf{v}|}{|\mathbf{v}|} \tag{2.1.6.1}$$
$$= \frac{|(1,2,2)\times(2,3,4)|}{\sqrt{2^2+3^2+4^2}} \tag{2.1.6.2}$$
$$= \frac{\sqrt{2^2+0^2+(-1)^2}}{\sqrt{29}}. \tag{2.1.6.3}$$

Hence $\boxed{d = \sqrt{\frac{5}{29}}}$.

2.1.7 Find a unit vector perpendicular to the plane of triangle ABC, where $A(3,-1,2)$,

$B(1,-1,3)$ and $C(4,-3,1)$.

Solution: The two side vectors from A are

$$\overrightarrow{AB} = (-2,0,1), \qquad \overrightarrow{AC} = (1,-2,-1). \tag{2.1.7.1}$$

Their cross product is a normal vector to the plane:

$$\overrightarrow{AB} \times \overrightarrow{AC} = (2,-1,4), \tag{2.1.7.2}$$

$$|\overrightarrow{AB} \times \overrightarrow{AC}| = \sqrt{2^2+(-1)^2+4^2} = \sqrt{21}. \tag{2.1.7.3}$$

Therefore the two possible unit normals are

$$\boxed{\pm\frac{2\hat{i}-\hat{j}+4\hat{k}}{\sqrt{21}}}. \tag{2.1.7.4}$$

B. Source audit notes

The first two vector questions contain wording/notation defects in the source transcription and remain conditional as marked above. The complete source includes further topic sections and a duplicated plane question; those have not yet been fully migrated or solved in this working batch.